

Delivery Operations Dashboard

Power BI Case Study

Problem Statement :

SwiftRoute Logistics operates across 6 hubs in Texas with 55 drivers and 45 vehicles. The business needs to identify what's causing delivery delays, which hubs and drivers are underperforming, and whether aging vehicles are a reliability risk — so operations can be improved and customer satisfaction scores can be raised.

Insights :

- 1. Austin Hub is the weakest hub** Austin has the lowest on-time rate at **77.9%** — 2.7 points below the top performer El Paso (80.6%). It also processes 4,065 orders with only 220 capacity, meaning it's consistently overloaded.
- 2. One driver is responsible for a disproportionate number of delays** Charles Thompson has **324 delays** — nearly 20% more than the #2 driver Karen Rodriguez (274). These two alone account for a significant chunk of total delays and need immediate coaching or reassignment review.
- 3. Experience does NOT clearly predict performance** Looking at the scatter, drivers with 1–2 years experience have ratings ranging from 1 to 5. Same story at 7–10 years. This means experience alone isn't the issue — training quality or route assignment likely matters more.
- 4. Older vehicles break down significantly more** Vehicles aged **5+ years average 18–25 breakdowns**, while vehicles under 3 years average only 5–10. FT-010 (5 yrs, 28 breakdowns) and FT-002 (5.2 yrs, 25 breakdowns) are critical risks. 12 out of 45 vehicles are currently in maintenance.
- 5. Vans carry the entire operation** Vans handle **17,480 orders out of 27,979 total — that's 62% of all deliveries**. Trucks handle 23%. If van availability drops, operations collapse. Yet breakdowns are spread across all types.
- 6. Dallas and Houston hubs are overloaded relative to capacity** Dallas processed **7,345 orders** with a capacity of only 250/day. Houston processed **6,875** with 380/day capacity. These two hubs alone handle 51% of all orders — a single disruption there affects half the network.
- 7. Road Construction and Vehicle Breakdown are the top 2 delay causes — both at 623 delays** These are tied at the top. Road construction is external (hard to control), but vehicle breakdown is **100% internal and preventable** with proper maintenance scheduling.

My Suggestion:

1. FT-002, FT-010, FT-020 continue to break down Retire or remove vehicles with 5+ years and 20+ breakdowns.
2. Austin Hub lowest on-time rate Redistribute orders from Austin to Fort Worth or San Antonio.
3. Charles Thompson – 324 delays Flag for performance review, compare route difficulty before concluding